

Architecture: Data model (Android Room + lava-api-go SQL)

Revision: 1 **Last modified:** 2026-06-08T00:00:00Z **Classification:** project-specific

Two independent persistence layers. Every claim below is read from real schema files (§11.4.6 no-guessing).

1. Android client — Room (SQLite), lava.database.AppDatabase, schema v11

Source of truth: core/database/schemas/lava.database.AppDatabase/11.json (Room-generated; checked in). 20 tables in v11:

Group	Tables
Search & history	Search, Suggest, HistoryTopic, FavoriteSearch, FavoriteTopic, Bookmark
Forum	ForumCategoryEntity, ForumMetadata, forum_provider_selections, search_provider_selections
Endpoints / discovery	Endpoint
Provider config & credentials	provider_configs, provider_credentials, provider_credential_binding, provider_sync_toggle, credentials_entry
Cloned providers / mirrors	cloned_provider, tracker_mirror_health, tracker_mirror_user
Sync	sync_outbox

Schema history: 5.json → 10.json → 11.json (the v10→v11 migration added the use_anonymous column to provider config — the fix from the prior completeness cycle). Migrations are Room Migration objects in core/database; schemas are validated against the checked-in JSON.

erDiagram

```
    provider_configs ||--o{ provider_credential_binding : "binds"
    provider_credentials ||--o{ provider_credential_binding :
"used-by"
    provider_configs ||--o{ provider_sync_toggle : "sync flag"
    cloned_provider ||--o{ provider_configs : "clone of"
    Search ||--o{ search_provider_selections : "scoped to"
```

```
ForumCategoryEntity ||--o{ forum_provider_selections : "scoped
to"
    tracker_mirror_health }o--|| Endpoint : "health of"
```

(Relations inferred from table names + the provider-config feature; column-level FKs are documented in 11.json — read it for exact columns/indices.)

2. lava-api-go — SQL (Postgres or SQLite), schema `lava_api`

Storage is **pluggable** (`lava-api-go/internal/storage/{factory,sqlite}.go` — SQLite for local/dev, Postgres for the deployed stack). DDL source of truth: `lava-api-go/migrations/*.up.sql` (golang-migrate). Real tables (verbatim from the migration files):

lava_api.response_cache (0001 + realigned 0005 — must match submodules/cache/pkg/postgres's (cache_key, value, expires_at) INSERT shape; a 2.0.0 mismatch silently cached nothing — forensic note in 0001):

```
CREATE TABLE IF NOT EXISTS lava_api.response_cache (
    cache_key    TEXT PRIMARY KEY,
    value        BYTEA NOT NULL,
    expires_at   TIMESTAMPTZ
);
```

lava_api.request_audit (0002):

```
CREATE TABLE IF NOT EXISTS lava_api.request_audit (
    id BIGSERIAL PRIMARY KEY, received_at TIMESTAMPTZ NOT NULL
    DEFAULT now(),
    method TEXT NOT NULL, path TEXT NOT NULL, query TEXT,
    client_ip INET,
    auth_realm_hash TEXT, upstream_status SMALLINT, upstream_ms
    INTEGER,
    cache_outcome TEXT NOT NULL, bytes_out INTEGER
);
```

lava_api.rate_limit_bucket (0003) — token-bucket per (client_ip, route_class):

```
CREATE TABLE IF NOT EXISTS lava_api.rate_limit_bucket (
    client_ip INET NOT NULL, route_class TEXT NOT NULL,
    tokens DOUBLE PRECISION NOT NULL, last_refill_at TIMESTAMPTZ
    NOT NULL,
    PRIMARY KEY (client_ip, route_class)
);
```

lava_api.login_attempt (0004) — usernames stored only as username_hash (§6.H):

```
CREATE TABLE IF NOT EXISTS lava_api.login_attempt (
    id BIGSERIAL PRIMARY KEY, client_ip INET NOT NULL,
    username_hash TEXT NOT NULL,
```

```

        succeeded BOOLEAN NOT NULL, attempted_at TIMESTAMPTZ NOT NULL
        DEFAULT now()
    );

```

lava_api.provider_credentials (0006) — secrets stored **encrypted at the application layer** before insert (encrypted_password/encrypted_token/encrypted_api_key/encrypted_api_secret); plaintext never persisted (§6.H):

```

CREATE TABLE IF NOT EXISTS lava_api.provider_credentials (
    provider_id TEXT PRIMARY KEY, auth_type TEXT NOT NULL DEFAULT
    'none',
    username TEXT, encrypted_password TEXT, encrypted_token TEXT,
    encrypted_api_key TEXT, encrypted_api_secret TEXT,
    cookie_value TEXT,
    expires_at TIMESTAMPTZ, is_active BOOLEAN NOT NULL DEFAULT
    true,
    last_used_at TIMESTAMPTZ, created_at TIMESTAMPTZ NOT NULL
    DEFAULT now(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

lava_api.provider_configs (0007) — per-provider timeouts / mirror / capability toggles (timeout_ms, preferred_mirror_url, is_enabled, search_enabled, browse_enabled, ...; read the full file for the complete column set).

Migration 0005 is intentionally destructive on response_cache (a response cache repopulates on demand; request_audit is preserved separately) — see the file header.

3. Notes

- The two layers are **independent**: the Android Room DB is the app's local state; lava_api is the server's cache/audit/rate-limit/credential store. They are not replicated into each other.
- provider_credentials exists in **both** layers with different intent: Room stores the user's local credential entries (device-side); lava_api.provider_credentials stores server-side encrypted creds for the API's own multi-provider auth.
- **UNCONFIRMED**: exact Room column types/indices per table (documented in 11.json, not re-typed here); the Jackett sidecar adds no schema (it owns its own /config store outside lava_api).

Sources verified

- core/database/schemas/lava.database.AppDatabase/11.json (20 tables, v11).
- lava-api-go/migrations/000{1..7}*.up.sql (verbatim DDL above).
- lava-api-go/internal/storage/{factory,sqlite}.go (pluggable SQLite/Postgres).